

BDA Practical No.2

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Step 1:-Type the command given below:-

```
stop-all.sh
```

Step 2:-Now type:-

```
hdfs namenode -format
```

Step 3:-Now type:-

```
start-dfs.sh
```

Step 4:-Type:-

```
start-yarn.sh
```

Note:-If you face any errors in Step No.3 and 4 then refer BDA Practical No.1 pdf by Aditya Nikam

Step 5:-Type:-

```
jps
```

Step 6:-Now make directory named “matrix” by typing the below command:-

```
mkdir matrix
```

Step 7:-Go to directory named “matrix”

```
cd matrix
```

Step 8:-Now type the command given below:-

```
nano mapper.py
```

And now paste the code given below in the mapper.py file:-

```
#!/usr/bin/env python3
```

```
import sys
```

```
for line in sys.stdin:
```

```
    # Removed the extra spaces and added the missing '='
```

```
    tokens = line.strip().split(',')
```

```

if tokens[0] == 'A':
    for k in range(0, 2): # assuming output matrix has 2 columns
        # Fixed spacing
        print(f"{tokens[1]},{k}\tA,{tokens[2]},{tokens[3]}")
elif tokens[0] == 'B':
    for i in range(0, 2): # assuming output matrix has 2 rows
        # Added missing closing bracket for tokens[1]
        print(f"{i},{tokens[2]}\tB,{tokens[1]},{tokens[3]}")

```

Now press Ctrl + O, then press Enter and then press Ctrl + X

Step 9:- Now type the command given below:-

```
nano reducer.py
```

And now paste the code given below in the reducer.py file:-

```

#!/usr/bin/env python3

import sys

from collections import defaultdict

# Fixed variable naming (removed spaces)

current_key = None

A_vals = defaultdict(float)

B_vals = defaultdict(float)

for line in sys.stdin:

    # Added missing '='

    key, value = line.strip().split('\t')

    matrix, index, val = value.split(',')

    if current_key and current_key != key:

        # Added missing '=' and '*' operator

```

```

    result = sum(A_vals[k] * B_vals[k] for k in A_vals if k in B_vals)

    print(f"{current_key}\t{result}")

    # Fixed variable names

    A_vals.clear()

    B_vals.clear()

    current_key = key

    if matrix == 'A':

        A_vals[index] = float(val)

    else:

        # Fixed fragmented lines and missing assignment

        B_vals[index] = float(val)

# Output the last key

if current_key:

    # Fixed typos ('fork', 'kin') and missing operators

    result = sum(A_vals[k] * B_vals[k] for k in A_vals if k in B_vals)

    print(f"{current_key}\t{result}")

```

Now press Ctrl + O, then press Enter and then press Ctrl + X

Step 10:- Now type the command given below:-

```
chmod +x mapper.py
```

Step 11:- Now type:-

```
chmod +x reducer.py
```

Step 12:- Now type:-

```
hadoop fs -mkdir -p /input
```

Step 13:- Now type the below command to create a.csv file:-

```
cat <<EOF > a.csv
```

```
A,0,0,1
```

A,0,1,2

A,1,0,3

A,1,1,4

EOF

Step 14:- Now type the below command to create b.csv file:-

```
cat <<EOF > b.csv
```

B,0,0,5

B,1,0,6

B,0,1,7

B,1,1,8

EOF

Step 15:-Now type the below command:-

```
ls -l *.csv
```

The output will be like below:-

```
-rw-r--r-- 1 aditya aditya 32 May 9 09:26 a.csv
```

```
-rw-r--r-- 1 aditya aditya 32 May 9 09:27 b.csv
```

This means that files “a.csv” and “b.csv” were created

Step 16:-Now type:-

```
hadoop fs -put a.csv /input/a.csv
```

Step 17:-Type:-

```
hadoop fs -put b.csv /input/b.csv
```

Step 18:-Type the command given below:-

```
find /usr /opt ~ -name "hadoop-streaming*.jar" 2>/dev/null
```

Output will be something like below:-

```
/usr/local/hadoop/share/hadoop/tools/sources/hadoop-streaming-3.3.6-test-sources.jar  
/usr/local/hadoop/share/hadoop/tools/sources/hadoop-streaming-3.3.6-sources.jar  
/usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar
```

So see here you will get the path and version of hadoop streaming like see in the above output /usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar is hadoop streaming version and path. We will need this path in step 8 (Don't see/copy test_sources.jar or sources.jar file remember that). And after pasting the path ensure that you type forward slash at the end “\” in the first line of the above command as you can see.

Step 19:-Now type:-

```
hadoop jar /usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar \  
  
-files mapper.py, reducer.py \  
  
-mapper ./mapper.py \  
  
-reducer ./reducer.py \  
  
-input /input \  
  
-output /output_matrix
```

You will see a lot of "INFO" text rapidly scrolling on your screen. Wait until it finishes completely.

See here in this command given above the first line has will have the path and version number of hadoop streaming we got from Step No.17 so we that pasted that path after hadoop jar in the first line of the above command so your path may be little different depending on version number note this

Also ensure that you have named files as “mapper.py” and “reducer.py” if the names are different then replace the names in the above command otherwise above command will give error.

Step 19:-Now type the below command:-

```
hadoop fs -cat /output_matrix/part-00000
```

You will see the output like below:-

0,0 17.0

0,1 23.0

1,0 39.0

1,1 53.0

If the output is like above means your practical is completed